

Jenkins Learning Roadmap

Beginner Level (Basics & Foundations)

- Introduction to Jenkins: What is Jenkins, CI/CD concepts, Jenkins architecture.
- Installation on Windows/Linux/Docker and UI Overview.
- Freestyle Jobs: Creating, configuring, SCM integration, build triggers.
- Pipeline Basics: Declarative vs Scripted, basic syntax (pipeline, agent, stages, steps).
- Run a simple Build → Test → Deploy pipeline.

Intermediate Level (Automation & Pipelines)

- Pipeline Features: environment variables, parameters (String, Choice, Boolean, Password, File).
- Post actions: success, failure, always, unstable.
- Control Flow: when, parallel, timeout, retry, input steps.
- SCM & Build Tools: GitHub/GitLab PR pipelines, Maven, Gradle, npm, MSBuild, JUnit test reports.
- Notifications & Reporting: Email, Slack/MS Teams, JUnit & HTML reports.

Advanced Level (Security, Scaling & Integrations)

- Credentials & Secrets: Credentials Manager, withCredentials usage, storing API keys, SSH keys.
- Distributed Builds: Master-Agent setup, labels, Docker agents, Kubernetes dynamic agents.
- Plugins & Integrations: SonarQube, Nexus/Artifactory, Docker, Helm, Kubernetes.
- Security & Access Control: RBAC, securing Jenkins, backup & restore.

Expert Level (CI/CD Mastery & DevOps)

- Advanced Pipelines: Multi-branch, Matrix builds, fast-fail parallel pipelines, Shared Libraries.
- Monitoring & Observability: Prometheus, Grafana, health checks, alerts.
- Enterprise CI/CD: GitOps, Jenkins + Ansible, Jenkins + Terraform, Jenkins + Wazuh for DevSecOps.
- Real-World CI/CD Projects: Build → Test → Scan → Docker → Push → Deploy (Kubernetes/IIS/Nginx).
- Deployment Strategies: Blue-Green & Canary deployments, Cloud integrations (AWS, Azure, GCP).